Chapter 16. Service Quality Metrics and SLAs

Service-level agreements (SLAs) are a focal point of negotiations, contract terms, legal obligations, and runtime metrics and measurements. SLAs formalize the guarantees put forth by cloud providers, and correspondingly influence or determine the pricing models and payment terms. SLAs set cloud consumer expectations and are integral to how organizations build business automation around the utilization of cloud-based IT resources.

The guarantees made by a cloud provider to a cloud consumer are often carried forward, in that the same guarantees are made by the cloud consumer organization to its clients, business partners, or whomever will be relying on the services and solutions hosted by the cloud provider. It is therefore crucial for SLAs and related service quality metrics to be understood and aligned in support of the cloud consumer's business requirements, while also ensuring that the guarantees can, in fact, be realistically fulfilled consistently and reliably by the cloud provider. The latter consideration is especially relevant for cloud providers that host shared IT resources for high volumes of cloud consumers, each of which will have been issued its own SLA guarantees.

16.1. Service Quality Metrics

SLAs issued by cloud providers are human-readable documents that describe quality-of-service (QoS) features, guarantees, and limitations of one or more cloud-based IT resources.

SLAs use service quality metrics to express measurable QoS characteristics.

For example:

- *Availability* – up-time, outages, service duration
- *Reliability* – minimum time between failures, guaranteed rate of successful responses
- *Performance* – capacity, response time, and delivery time guarantees
- *Scalability* – capacity fluctuation and responsiveness guarantees
- *Resiliency* – mean-time to switchover and recovery

SLA management systems use these metrics to perform periodic measurements that verify compliance with SLA guarantees, in addition to collecting SLA-related data for various types of statistical analyses.

Each service quality metric is ideally defined using the following characteristics:

- *Quantifiable* – The unit of measure is clearly set, absolute, and appropriate so that the metric can be based on quantitative measurements.
- *Repeatable* – The methods of measuring the metric need to yield identical results when repeated under identical conditions.
- *Comparable* – The units of measure used by a metric need to be standardized and comparable. For example, a service quality metric cannot measure smaller quantities of data in bits and larger quantities in bytes.
- *Easily Obtainable* – The metric needs to be based on a non-proprietary, common form of measurement that can be easily obtained and understood by cloud consumers.

The upcoming sections provide a series of common service quality metrics, each of which is documented with description, unit of measure, measurement frequency, and applicable cloud delivery model values, as well as a brief example.

Service Availability Metrics

Availability Rate Metric

The overall availability of an IT resource is usually expressed as a percentage of up-time. For example, an IT resource that is always available will have an up-time of 100%.

- *Description* – percentage of service up-time
- *Measurement* – total up-time / total time
- *Frequency* – weekly, monthly, yearly
- *Cloud Delivery Model* – IaaS, PaaS, SaaS
- *Example* – minimum 99.5% up-time

Availability rates are calculated cumulatively, meaning that unavailability periods are combined in order to compute the total downtime (Table 16.1).

Availability (%)	Downtime/Week (Seconds)	Downtime/Month (Seconds)	Downtime/Year (Seconds)
99.5	3024	216	158112
99.8	1210	5174	63072
99.9	606	2592	31536
99.95	302	1294	15768
99.99	60.6	259.2	3154
99.999	6.05	25.9	316.6
99.9999	0.605	2.59	31.5

Table 16.1 Sample availability rates measured in units of seconds.

Outage Duration Metric

This service quality metric is used to define both maximum and average continuous outage service-level targets.

- *Description* – duration of a single outage
- *Measurement* – date/time of outage end – date/time of outage start
- *Frequency* – per event
- *Cloud Delivery Model* – IaaS, PaaS, SaaS
- *Example* – 1 hour maximum, 15 minute average

Note

In addition to being quantitatively measured, availability can be described qualitatively using terms such as high-availability (HA), which is used to label an IT resource with exceptionally low downtime usually due to underlying resource replication and/or clustering infrastructure.

Service Reliability Metrics

A characteristic closely related to availability, reliability is the probability that an IT resource can perform its intended function under pre-defined conditions without experiencing failure. Reliability focuses on how often the service performs as expected, which requires the service to remain in an operational and available state. Certain reliability metrics only consider runtime errors and exception conditions as failures, which are commonly measured only when the IT resource is available.

Mean-Time Between Failures (MTBF) Metric

- *Description* – expected time between consecutive service failures
- *Measurement* – Σ normal operational period duration / number of failures

- *Frequency* – monthly, yearly
- *Cloud Delivery Model* – IaaS, PaaS
- *Example* – 90 day average

Reliability Rate Metric

Overall reliability is more complicated to measure and is usually defined by a reliability rate that represents the percentage of successful service outcomes. This metric measures the effects of non-fatal errors and failures that occur during up-time periods. For example, an IT resource's reliability is 100% if it has performed as expected every time it is invoked, but only 80% if it fails to perform every fifth time.

- *Description* – percentage of successful service outcomes under pre-defined conditions
- *Measurement* – total number of successful responses / total number of requests
- *Frequency* – weekly, monthly, yearly
- *Cloud Delivery Model* – SaaS
- *Example* – minimum 99.5%

Service Performance Metrics

Service performance refers to the ability on an IT resource to carry out its functions within expected parameters. This quality is measured using service capacity metrics, each of which focuses on a related measurable characteristic of IT resource capacity. A set of common performance capacity metrics is provided in this section. Note that different metrics may apply, depending on the type of IT resource being measured.

Network Capacity Metric

- *Description* – measurable characteristics of network capacity
- *Measurement* – bandwidth / throughput in bits per second
- *Frequency* – continuous
- *Cloud Delivery Model* – IaaS, PaaS, SaaS
- *Example* – 10 MB per second

Storage Device Capacity Metric

- *Description* – measurable characteristics of storage device capacity
- *Measurement* – storage size in GB
- *Frequency* – continuous
- *Cloud Delivery Model* – IaaS, PaaS, SaaS
- *Example* – 80 GB of storage

Server Capacity Metric

- *Description* – measurable characteristics of server capacity
- *Measurement* – number of CPUs, CPU frequency in GHz, RAM size in GB, storage size in GB

- *Frequency* – continuous
- *Cloud Delivery Model* – IaaS, PaaS
- *Example* – 1 core at 1.7 GHz, 16 GB of RAM, 80 GB of storage

Web Application Capacity Metric

- *Description* – measurable characteristics of Web application capacity
- *Measurement* – rate of requests per minute
- *Frequency* – continuous
- *Cloud Delivery Model* – SaaS
- *Example* – maximum 100,000 requests per minute

Instance Starting Time Metric

- *Description* – length of time required to initialize a new instance
- *Measurement* – date/time of instance up – date/time of start request
- *Frequency* – per event
- *Cloud Delivery Model* – IaaS, PaaS
- *Example* – 5 minute maximum, 3 minute average

Response Time Metric

- *Description* – time required to perform synchronous operation
- *Measurement* – (date/time of request – date/time of response) / total number of requests
- *Frequency* – daily, weekly, monthly
- *Cloud Delivery Model* – SaaS
- *Example* – 5 millisecond average

Completion Time Metric

- *Description* – time required to complete an asynchronous task
- *Measurement* – (date of request – date of response) / total number of requests
- *Frequency* – daily, weekly, monthly
- *Cloud Delivery Model* – PaaS, SaaS
- *Example* – 1 second average

Service Scalability Metrics

Service scalability metrics are related to IT resource elasticity capacity, which is related to the maximum capacity that an IT resource can achieve, as well as measurements of its ability to adapt to workload fluctuations. For example, a server can be scaled up to a maximum of 128 CPU cores and 512 GB of RAM, or scaled out to a maximum of 16 load-balanced replicated instances.

The following metrics help determine whether dynamic service demands will be met proactively or reactively, as well as the impacts of manual or automated IT resource allocation processes.

Storage Scalability (Horizontal) Metric

- *Description* – permissible storage device capacity changes in response to increased workloads
- *Measurement* – storage size in GB
- *Frequency* – continuous
- *Cloud Delivery Model* – IaaS, PaaS, SaaS
- *Example* – 1,000 GB maximum (automated scaling)

Server Scalability (Horizontal) Metric

- *Description* – permissible server capacity changes in response to increased workloads
- *Measurement* – number of virtual servers in resource pool
- *Frequency* – continuous
- *Cloud Delivery Model* – IaaS, PaaS
- *Example* – 1 virtual server minimum, 10 virtual server maximum (automated scaling)

Server Scalability (Vertical) Metric

- *Description* – permissible server capacity fluctuations in response to workload fluctuations
- *Measurement* – number of CPUs, RAM size in GB
- *Frequency* – continuous
- *Cloud Delivery Model* – IaaS, PaaS
- *Example* – 512 core maximum, 512 GB of RAM

Service Resiliency Metrics

The ability of an IT resource to recover from operational disturbances is often measured using service resiliency metrics. When resiliency is described within or in relation to SLA resiliency guarantees, it is often based on redundant implementations and resource replication over different physical locations, as well as various disaster recovery systems.

The type of cloud delivery model determines how resiliency is implemented and measured. For example, the physical locations of replicated virtual servers that are implementing resilient cloud services can be explicitly expressed in the SLAs for IaaS environments, while being implicitly expressed for the corresponding PaaS and SaaS environments.

Resiliency metrics can be applied in three different phases to address the challenges and events that can threaten the regular level of a service:

- *Design Phase* – Metrics that measure how prepared systems and services are to cope with challenges.
- *Operational Phase* – Metrics that measure the difference in service levels before, during, and after a downtime event or service outage, which are further qualified by availability, reliability, performance, and scalability metrics.

- *Recovery Phase* – Metrics that measure the rate at which an IT resource recovers from downtime, such as the meantime for a system to log an outage and switchover to a new virtual server.

Two common metrics related to measuring resiliency are as follows:

Mean-Time to Switchover (MTSO) Metric

- *Description* – the time expected to complete a switchover from a severe failure to a replicated instance in a different geographical area
- *Measurement* – (date/time of switchover completion – date/time of failure) / total number of failures
- *Frequency* – monthly, yearly
- *Cloud Delivery Model* – IaaS, PaaS, SaaS
- *Example* – 10 minute average

Mean-Time System Recovery (MTSR) Metric

- *Description* – time expected for a resilient system to perform a complete recovery from a severe failure
- *Measurement* – (date/time of recovery – date/time of failure) / total number of failures
- *Frequency* – monthly, yearly
- *Cloud Delivery Model* – IaaS, PaaS, SaaS
- *Example* – 120 minute average

16.2. Case Study Example

After suffering a cloud outage that made their Web portal unavailable for about an hour, Innovartus decides to thoroughly review the terms and conditions of their SLA. They begin by researching the cloud provider's availability guarantees, which prove to be ambiguous because they do not clearly state which events in the cloud provider's SLA management system are classified as "downtime." Innovartus also discovers that the SLA lacks reliability and resilience metrics, which had become essential to their cloud service operations.

In preparation for a renegotiation of the SLA terms with the cloud provider, Innovartus decides to compile a list of additional requirements and guarantee stipulations:

- The availability rate needs to be described in greater detail to enable more effective management of service availability conditions.
- Technical data that supports service operations models needs to be included in order to ensure that the operation of select critical services remains fault-tolerant and resilient.
- Additional metrics that assist in service quality assessment need to be included.
- Any events that are to be excluded from what is measured with availability metrics need to be clearly defined.

After several conversations with the cloud provider sales represenatative, Innovartus is offered a revised SLA with the following additions:

- The method by which the availability of cloud services are to be measured, in addition to any supporting IT resources on which ATN core processes depend.
- Inclusion of a set of reliability and performance metrics approved by Innovartus.

Six months later, Innovartus performs another SLA metrics assessment and compares the newly generated values with ones that were generated prior to the SLA improvements (Table 16.2).

SLA Metrics	Statistics of Previous SLA	Statistics of Revised SLA
Average Availability	98.10%	99.98%
High-Availability Model	Cold-Standby	Hot-Standby
Average Service Quality **based on customer satisfaction surveys.*	52%	70%

Table 16.2 The evolution of Innovartus' SLA evaluation, as monitored by their cloud resource administrators.

~~16.3. SLA Guidelines~~

This section provides a number of best practices and recommendations for working with SLAs, the majority of which are applicable to cloud consumers:

- *Mapping Business Cases to SLAs* – It can be helpful to identify the necessary QoS requirements for a given automation solution and to then concretely link them to the guarantees expressed in the SLAs for IT resources responsible for carrying out the automation. This can avoid situations where SLAs are inadvertently misaligned or perhaps unreasonably deviate in their guarantees, subsequent to IT resource usage.
- *Working with Cloud and On-Premise SLAs* – Due to the vast infrastructure available to support IT resources in public clouds, the QoS guarantees issued in SLAs for cloud-based IT resources are generally superior to those provided for on-premise IT resources. This variance needs to be understood, especially when building hybrid distributed solutions that utilize both on on-premise and cloud-based services or when incorporating cross-environment technology architectures, such as cloud bursting.
- *Understanding the Scope of an SLA* – Cloud environments are comprised of many supporting architectural and infrastructure layers upon which IT resources reside and are integrated. It is important to acknowledge the extent to which a given IT resource guarantee applies. For example, an SLA may be limited to the IT resource implementation but not its underlying hosting environment.
- *Understanding the Scope of SLA Monitoring* – SLAs need to specify where monitoring is performed and where measurements are calculated, primarily in relation to the cloud's firewall. For example, monitoring within the cloud firewall is not always advantageous or relevant to the cloud consumer's required QoS guarantees. Even the most efficient firewalls have a measurable degree of influence on performance and can further present a point of failure.

- *Documenting Guarantees at Appropriate Granularity* – SLA templates used by cloud providers sometimes define guarantees in broad terms. If a cloud consumer has specific requirements, the corresponding level of detail should be used to describe the guarantees. For example, if data replication needs to take place across particular geographic locations, then these need to be specified directly within the SLA.
- *Defining Penalties for Non-Compliance* – If a cloud provider is unable to follow through on the QoS guarantees promised within the SLAs, recourse can be formally documented in terms of compensation, penalties, reimbursements, or otherwise.
- *Incorporating Non-Measurable Requirements* – Some guarantees cannot be easily measured using service quality metrics, but are relevant to QoS nonetheless, and should therefore still be documented within the SLA. For example, a cloud consumer may have specific security and privacy requirements for data hosted by the cloud provider that can be addressed by assurances in the SLA for the cloud storage device being leased.
- *Disclosure of Compliance Verification and Management* – Cloud providers are often responsible for monitoring IT resources to ensure compliance with their own SLAs. In this case, the SLAs themselves should state what tools and practices are being used to carry out the compliance checking process, in addition to any legal-related auditing that may be occurring.
- *Inclusion of Specific Metric Formulas* – Some cloud providers do not mention common SLA metrics or the metrics-related calculations in their SLAs, instead focusing on service-level descriptions that highlight the use of best practices and customer support. Metrics being used to measure SLAs should be part of the SLA document, including the formulas and calculations that the metrics are based upon.
- *Considering Independent SLA Monitoring* – Although cloud providers will often have sophisticated SLA management systems and SLA monitors, it may be in the best interest of a cloud consumer to hire a third-party organization to perform independent monitoring as well, especially if there are suspicions that SLA guarantees are not always being met by the cloud provider (despite the results shown on periodically issued monitoring reports).
- *Archiving SLA Data* – The SLA-related statistics collected by SLA monitors are commonly stored and archived by the cloud provider for future reporting purposes. If a cloud provider intends to keep SLA data specific to a cloud consumer even after the cloud consumer no longer continues its business relationship with the cloud provider, then this should be disclosed. The cloud consumer may have data privacy requirements that disallow the unauthorized storage of this type of information. Similarly, during and after a cloud consumer's engagement with a cloud provider, it may want to keep a copy of historical SLA-related data as well. It may be especially useful for comparing cloud providers in the future.
- *Disclosing Cross-Cloud Dependencies* – Cloud providers may be leasing IT resources from other cloud providers, which results in a loss of control over the guarantees they are able to make to cloud consumers. Although a cloud provider will rely on the SLA assurances made to it by other cloud providers, the cloud consumer may want disclosure of the fact that the IT resources it is leasing may have dependencies beyond the environment of the cloud provider organization that it is leasing them

from.

16.4. Case Study Example

DTGOV begins its SLA template authoring process by working with a legal advisory team that has been adamant about an approach whereby cloud consumers are presented with an online Web page outlining the SLA guarantees, along with a "click-once-to-accept" button. The default agreement contains extensive limitations to DTGOV's liability in relation to possible SLA non-compliance, as follows:

- The SLA defines guarantees only for service availability.
- Service availability is defined for all of the cloud services simultaneously.
- Service availability metrics are loosely defined to establish a level of flexibility regarding unexpected outages.
- The terms and conditions are linked to the Cloud Services Customer Agreement, which is accepted implicitly by all of the cloud consumers that use the self-service portal.
- Extended periods of unavailability are to be recompensed by monetary "service credits," which are to be discounted on future invoices and have no actual monetary value.

Provided here are key excerpts from DTGOV's SLA template:

Scope and Applicability

This Service Level Agreement ("SLA") establishes the service quality parameters that are to be applied to the use of DTGOV's cloud services ("DTGOV cloud"), and is part of the DTGOV Cloud Services Customer Agreement ("DTGOV Cloud Agreement").

The terms and conditions specified in this agreement apply solely to virtual server and cloud storage device services, herein called "Covered Services." This SLA applies separately to each cloud consumer ("Consumer") that is using the DTGOV Cloud. DTGOV reserves the right to change the terms of this SLA in accordance with the DTGOV Cloud Agreement at any time.

Service Quality Guarantees

The Covered Services will be operational and available to Consumers at least 99.95% of the time in any calendar month. If DTGOV does not meet this SLA requirement while the Consumer succeeds in meeting its SLA obligations, the Consumer will be eligible to receive Financial Credits as compensation. This SLA states the Consumer's exclusive right to compensation for any failure on DTGOV's part to fulfill the SLA requirements.

Definitions

The following definitions are to be applied to DTGOV's SLA:

- "Unavailability" is defined as the entirety of the Consumer's running instances as having no external connectivity for a duration that is at least five consecutive minutes in length, during which the Consumer is unable to launch commands against the remote administration system through either the Web application or Web service API.

- “Downtime Period” is defined as a period of five or more consecutive minutes of the service remaining in a state of Unavailability. Periods of “Intermittent Downtime” that are less than five minutes long do not count towards Downtime Periods.
- “Monthly Up-time Percentage” (MUP) is calculated as: (total number of minutes in a month – total number of downtime period minutes in a month) / (total number of minutes in a month)
- “Financial Credit” is defined as the percentage of the monthly invoice total that is credited towards future monthly invoices of the Consumer, which is calculated as follows:

  99.00% < MUP % < 99.95% – 10% of the monthly invoice is credited in favor of the Consumer’s invoice

  89.00% < MUP % < 99.00% – 30% of the monthly invoice is credited in favor of the Consumer’s invoice

  MUP % < 89.00% – 100% of the monthly invoice is credited in favor of the Consumer’s invoice

Usage of Financial Credits

The MUP for each billing period is to be displayed on each monthly invoice. The Consumer is to submit a request for Financial Credit in order to be eligible to redeem Financial Credits. For that purpose, the Consumer is to notify DTGOV within thirty days from the time the Consumer receives the invoice that states the MUP beneath the defined SLA. Notification is to be sent to DTGOV via e-mail. Failure to comply with this requirement forfeits the Consumer’s right to the redemption of Financial Credits.

SLA Exclusions

The SLA does not apply to any of the following:

- Unavailability periods caused by factors that cannot be reasonably foreseen or prevented by DTGOV.
- Unavailability periods resulting from the malfunctioning of the Consumer’s software and/or hardware, third party software and/or hardware, or both.
- Unavailability periods resulting from abuse or detrimental behavior and actions that are in violation of the DTGOV Cloud Agreement.
- Consumers with overdue invoices or are otherwise not considered in good standing with DTGOV.